

1) Why do we maximize the log-likelihood, instead of the likelihood?

The likelihood function is a bunch of multiplications and log is an almost magical function that turns

- exponentiation into multiplication
- multiplication into addition
- addition is close to constants

2) Why do we negate the log-likelihood function?

Since $0 \leq \theta \leq 1$, it will be a fraction, and when exponentiating fractions, we get really small fractions. Floating point arithmetic gets weird for numbers that are around very large and very small powers of 10. The log function makes those numbers more reasonable, and specifically relating to fractional numbers, which θ will be, become negative. If our numbers are always negative, we can save memory/computation by simply changing our scale by negating it. Then we no longer have to store the negative flag, and we can get larger numbers from unsigned data types.

Once again we are rotating into a frame of reference that is convenient.

3) Suppose someone adds 5000 to the loss function

Will the optimum parameter values change?

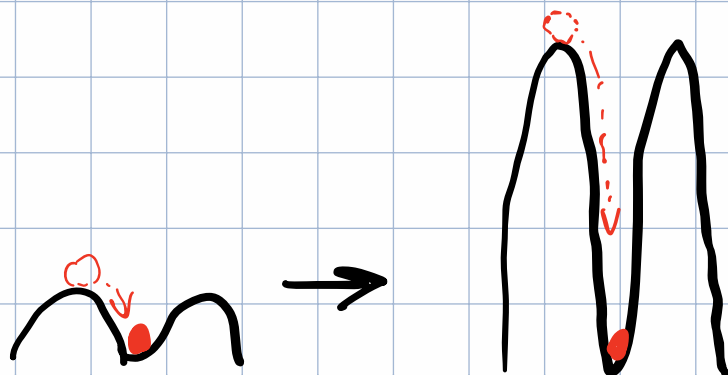
No, they won't change. They'll just be shifted, and we're just going to be re-centering anyway.

4) Suppose we multiply a loss function by 100. Will the optimum change? Will gradient descent behave exactly the same?

No, we're going to rescale that away, especially with us using the log on it which turns multiplication into addition (an offset).

Gradient descent should be unaffected. It doesn't matter what the relative height of the peaks will be. They'll be in the same place, and so will the minimums.

Raise the height of hills, and a ball will still roll to the same place.



5) Why do we use least squares when we assume Gaussian noise?

Our data is the signal plus the noise

$$y_i = \hat{y} + \epsilon_i \quad \text{so for any signal, } y_i,$$

$$\epsilon_i = y_i - \hat{y}$$

y_i can be represented by a matrix X , and our noise is just how far off we are from the center (our mean)

$$\tilde{X} = X - E(X)$$

inner product

$$\text{Our Cov}(X, y) = E[(X - E[X])(y - E[y])]$$

Which leads to our covariance matrix Σ , and our Likelihood being Σ^{-1}

Least squares is a natural consequence